No.	Status	Date
LNDE-RPT-20191001-027-DS-4-RA-101	Issued	2019-10-18

UT Thickness Data Sheet - 4-RA-101

Scope and Responsibilities

This data sheet provides the ultrasonic thickness measurement results for line 4-RA-101 located within the ATU-100. The objective is to establish a thickness re-baseline for the rich amine service line, consisting of A106 Gr.B piping with a nominal wall of 8.56 mm and a design pressure of 862 kPa(g) at 82 °C.

Trevor Nolan performed the ultrasonic thickness measurements. The technician is qualified to SNT-TC-1A UT Level I (2014) and UT Level II (2016-09) through Larkhall NDE Services Inc.

Technical Basis

All thickness measurements were conducted in accordance with ASTM E797. The inspection was performed under the requirements of QSCP-NDT-001 Rev 2, which includes specific mandates for insulation removal and reinstatement as part of the CUI program.

The data presented herein supports the findings detailed in the re-baseline survey report LNDE-RPT-20191001-027-R0, which covers 150 CMLs across the ATU-100.

Thickness Measurements and Analysis

Measurements for 4-RA-101 were taken at designated CMLs. The recorded wall thicknesses are 2.83 mm at 2-RF-102-04, 2.88 mm at 2-RF-102-05, 2.74 mm at 2-RF-102-06, and 2.82 mm at 2-RF-102-07.

Based on the retirement thickness (t-min) of 3.81 mm, all four measured locations are below the minimum allowable wall thickness. These readings are raised as NCRs.

Remaining Life and Intervals

The calculated remaining life for the identified CMLs is 4.5 years for 2-RF-102-04, 4.3 years for 2-RF-102-05, 2.9 years for 2-RF-102-06, and 3.2 years for 2-RF-102-07.

As a rich amine line, 4-RA-101 is categorized under a 3 year RBI interval. The next thickness reading must be scheduled at no more than half the remaining life, subject to the Class 2 maximum of 10 years.

Thickness readings, measured 2019-10-18

Line 6-SG-101 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-101-01	7.11	3.56	2016-02-12	6.17	5.93	0.050	0.065	36.3	2022-10-18	
6-SG-101-02	7.11	3.56	2016-02-12	5.76	5.51	0.065	0.068	28.7	2022-10-18	
6-SG-101-03	7.11	3.56	2016-02-12	5.87	5.56	0.064	0.084	23.7	2022-10-18	
6-SG-101-04	7.11	3.56	2016-02-12	5.58	5.34	0.074	0.065	24.0	2022-10-18	
6-SG-101-05	7.11	3.56	2016-02-12	5.38	5.04	0.089	0.092	16.0	2022-10-18	
6-SG-101-06	7.11	3.56	2016-02-12	5.88	5.56	0.064	0.087	23.0	2022-10-18	
6-SG-101-07	7.11	3.56	—	—	6.24	—	—	—	2022-10-18	
6-SG-101-08	7.11	3.56	2016-02-12	6.00	5.77	0.065	0.063	34.2	2022-10-18	

Line 6-SG-102 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-102-01	7.11	3.56	2016-02-12	5.42	5.05	0.089	0.101	14.8	2022-10-18	
6-SG-102-02	7.11	3.56	2016-02-12	5.43	5.10	0.085	0.090	17.2	2022-10-18	
6-SG-102-03	7.11	3.56	2016-02-12	5.22	4.92	0.090	0.082	15.1	2022-10-18	
6-SG-102-04	7.11	3.56	2016-02-12	5.18	4.90	0.089	0.076	15.1	2022-10-18	
6-SG-102-05	7.11	3.56	2016-02-12	6.19	6.07	0.045	0.033	>50	2022-10-18	
6-SG-102-06	7.11	3.56	2016-02-12	6.06	6.03	0.044	0.008	>50	2022-10-18	

Line 8-TG-101 | Treated gas | A106 Gr.B | t_nom 8.18 mm | t-min 4.06 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
8-TG-101-01	8.18	4.06	2016-02-12	7.76	7.63	0.024	0.035	>50	2025-10-18	
8-TG-101-02	8.18	4.06	2016-02-12	7.46	7.33	0.037	0.035	>50	2025-10-18	
8-TG-101-03	8.18	4.06	2016-02-12	7.54	7.37	0.036	0.046	>50	2025-10-18	
8-TG-101-04	8.18	4.06	2016-02-12	7.51	7.38	0.030	0.035	>50	2025-10-18	
8-TG-101-05	8.18	4.06	2016-02-12	7.73	7.57	0.028	0.043	>50	2025-10-18	
8-TG-101-06	8.18	4.06	—	—	7.75	—	—	—	2025-10-18	
8-TG-101-07	8.18	4.06	2016-02-12	7.39	7.17	0.060	0.060	>50	2025-10-18	

Line 4-LA-101 | Lean amine supply | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-101-01	6.02	3.18	2016-02-12	5.02	4.70	0.055	0.087	17.5	2025-10-18	
4-LA-101-02	6.02	3.18	2016-02-12	5.45	5.26	0.032	0.052	40.3	2025-10-18	
4-LA-101-03	6.02	3.18	2016-02-12	5.45	5.38	0.025	0.019	>50	2025-10-18	
4-LA-101-04	6.02	3.18	2016-02-12	5.48	5.29	0.033	0.052	40.9	2025-10-18	
4-LA-101-05	6.02	3.18	2016-02-12	5.54	5.38	0.027	0.043	>50	2025-10-18	
4-LA-101-06	6.02	3.18	2016-02-12	5.30	5.22	0.036	0.022	>50	2025-10-18	
4-LA-101-07	6.02	3.18	2016-02-12	4.87	4.86	0.048	0.003	35.1	2025-10-18	
4-LA-101-08	6.02	3.18	—	—	5.04	—	—	—	2025-10-18	
4-LA-101-09	6.02	3.18	2016-02-12	5.00	4.86	0.038	0.038	44.2	2025-10-18	

Line 4-LA-102 | Lean amine supply | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-102-01	6.02	3.18	2016-02-12	5.25	5.09	0.039	0.043	43.9	2025-10-18	
4-LA-102-02	6.02	3.18	2016-02-12	5.36	5.15	0.039	0.057	34.5	2025-10-18	
4-LA-102-03	6.02	3.18	2016-02-12	5.28	5.15	0.039	0.035	>50	2025-10-18	
4-LA-102-04	6.02	3.18	2016-02-12	4.90	4.65	0.057	0.068	21.6	2025-10-18	
4-LA-102-05	6.02	3.18	2016-02-12	5.34	5.23	0.033	0.030	>50	2025-10-18	
4-LA-102-06	6.02	3.18	2016-02-12	5.60	5.52	0.020	0.022	>50	2025-10-18	
4-LA-102-07	6.02	3.18	2016-02-12	5.40	5.32	0.029	0.022	>50	2025-10-18	
4-LA-102-08	6.02	3.18	2016-02-12	5.70	5.55	0.041	0.041	>50	2025-10-18	

Line 6-LA-103 | Lean amine header | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-LA-103-01	7.11	3.56	2016-02-12	6.44	6.40	0.026	0.011	>50	2025-10-18	
6-LA-103-02	7.11	3.56	2016-02-12	6.53	6.49	0.030	0.011	>50	2025-10-18	
6-LA-103-03	7.11	3.56	2016-02-12	6.28	6.23	0.036	0.014	>50	2025-10-18	
6-LA-103-04	7.11	3.56	2016-02-12	6.15	5.85	0.056	0.082	28.1	2025-10-18	
6-LA-103-05	7.11	3.56	2016-02-12	5.98	5.81	0.058	0.046	38.5	2025-10-18	
6-LA-103-06	7.11	3.56	2016-02-12	5.82	5.46	0.072	0.098	19.4	2025-10-18	
6-LA-103-07	7.11	3.56	2016-02-12	6.12	5.93	0.053	0.052	44.8	2025-10-18	
6-LA-103-08	7.11	3.56	2016-02-12	6.49	6.19	0.036	0.082	32.3	2025-10-18	
6-LA-103-09	7.11	3.56	—	—	6.16	—	—	—	2025-10-18	
6-LA-103-10	7.11	3.56	2016-02-12	6.30	6.00	0.082	0.082	29.9	2025-10-18	

Line 4-RA-101 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-101-01	8.56	3.81	2016-02-12	7.83	7.29	0.052	0.147	23.7	2022-10-18	
4-RA-101-02	8.56	3.81	2016-02-12	7.94	7.39	0.049	0.149	24.0	2022-10-18	
4-RA-101-03	8.56	3.81	2016-02-12	7.65	6.64	0.083	0.274	10.3	2022-10-18	

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-101-04	8.56	3.81	2016-02-12	8.00	7.54	0.046	0.125	29.8	2022-10-18	
4-RA-101-05	8.56	3.81	2016-02-12	7.82	7.18	0.060	0.174	19.4	2022-10-18	
4-RA-101-06	8.56	3.81	2016-02-12	7.87	6.94	0.191	0.253	12.4	2022-10-18	
4-RA-101-07	8.56	3.81	2016-02-12	8.14	7.68	0.039	0.125	31.0	2022-10-18	
4-RA-101-08	8.56	3.81	2016-02-12	7.31	6.45	0.094	0.234	11.3	2022-10-18	
4-RA-101-09	8.56	3.81	2016-02-12	8.13	7.68	0.038	0.122	31.6	2022-10-18	
4-RA-101-10	8.56	3.81	2016-02-12	8.02	7.55	0.047	0.128	29.3	2022-10-18	
4-RA-101-11	8.56	3.81	2016-02-12	7.60	6.66	0.086	0.255	11.2	2022-10-18	
4-RA-101-12	8.56	3.81	—	—	7.03	—	—	—	2022-10-18	

Line 4-RA-102 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-102-01	8.56	3.81	2016-02-12	8.02	7.58	0.041	0.120	31.5	2022-10-18	
4-RA-102-02	8.56	3.81	2016-02-12	7.64	6.96	0.068	0.185	17.0	2022-10-18	
4-RA-102-03	8.56	3.81	2016-02-12	7.55	7.05	0.064	0.136	23.8	2022-10-18	
4-RA-102-04	8.56	3.81	2016-02-12	7.72	7.04	0.064	0.185	17.5	2022-10-18	
4-RA-102-05	8.56	3.81	2016-02-12	7.67	7.06	0.063	0.166	19.6	2022-10-18	
4-RA-102-06	8.56	3.81	2016-02-12	7.73	7.15	0.058	0.158	21.2	2022-10-18	
4-RA-102-07	8.56	3.81	2016-02-12	7.78	7.30	0.054	0.130	26.8	2022-10-18	
4-RA-102-08	8.56	3.81	2016-02-12	7.35	6.40	0.089	0.258	10.0	2022-10-18	
4-RA-102-09	8.56	3.81	2016-02-12	7.54	6.78	0.075	0.207	14.4	2022-10-18	
4-RA-102-10	8.56	3.81	—	—	7.36	—	—	—	2022-10-18	
4-RA-102-11	8.56	3.81	2016-02-12	7.96	7.72	0.054	0.065	>50	2022-10-18	

Line 6-RA-103 | Rich amine header | A106 Gr.B | t_nom 10.97 mm | t-min 4.57 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-RA-103-01	10.97	4.57	2016-02-12	10.29	9.83	0.050	0.125	42.1	2022-10-18	
6-RA-103-02	10.97	4.57	2016-02-12	9.69	8.61	0.105	0.294	13.8	2022-10-18	
6-RA-103-03	10.97	4.57	2016-02-12	10.06	9.49	0.064	0.155	31.8	2022-10-18	
6-RA-103-04	10.97	4.57	2016-02-12	9.31	8.22	0.117	0.296	12.3	2022-10-18	
6-RA-103-05	10.97	4.57	2016-02-12	9.48	8.33	0.111	0.313	12.0	2022-10-18	
6-RA-103-06	10.97	4.57	2016-02-12	10.32	9.79	0.052	0.144	36.2	2022-10-18	
6-RA-103-07	10.97	4.57	2016-02-12	10.20	9.59	0.059	0.166	30.3	2022-10-18	
6-RA-103-08	10.97	4.57	2016-02-12	10.33	9.73	0.053	0.163	31.6	2022-10-18	
6-RA-103-09	10.97	4.57	2016-02-12	10.33	9.78	0.052	0.149	34.9	2022-10-18	
6-RA-103-10	10.97	4.57	2016-02-12	9.49	8.37	0.112	0.304	12.5	2022-10-18	
6-RA-103-11	10.97	4.57	2016-02-12	9.77	8.91	0.234	0.234	18.5	2022-10-18	
6-RA-103-12	10.97	4.57	—	—	9.52	—	—	—	2022-10-18	
6-RA-103-13	10.97	4.57	2016-02-12	10.15	9.55	0.104	0.163	30.5	2022-10-18	
6-RA-103-14	10.97	4.57	—	—	8.26	—	—	—	2022-10-18	

Line 3-RV-101 | Regenerator overhead | A106 Gr.B | t_nom 7.62 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RV-101-01	7.62	3.56	2016-02-12	6.34	6.05	0.065	0.079	31.6	2022-10-18	
3-RV-101-02	7.62	3.56	2016-02-12	6.31	5.97	0.068	0.092	26.1	2022-10-18	
3-RV-101-03	7.62	3.56	2016-02-12	5.63	5.24	0.097	0.106	15.9	2022-10-18	
3-RV-101-04	7.62	3.56	2016-02-12	5.88	5.58	0.085	0.082	23.7	2022-10-18	
3-RV-101-05	7.62	3.56	2016-02-12	6.36	6.11	0.066	0.068	37.5	2022-10-18	
3-RV-101-06	7.62	3.56	2016-02-12	5.96	5.55	0.086	0.111	17.9	2022-10-18	
3-RV-101-07	7.62	3.56	2016-02-12	5.54	5.07	0.124	0.128	11.8	2022-10-18	

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RV-101-08	7.62	3.56	2016-02-12	6.31	5.86	0.093	0.122	18.8	2022-10-18	

Line 2-RV-102 | Regenerator overhead | A106 Gr.B | t_nom 5.54 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RV-102-01	5.54	2.79	2016-02-12	4.35	4.16	0.058	0.052	23.6	2022-10-18	
2-RV-102-02	5.54	2.79	2016-02-12	4.01	3.86	0.073	0.041	14.6	2022-10-18	
2-RV-102-03	5.54	2.79	2016-02-12	4.64	4.37	0.047	0.073	21.5	2022-10-18	
2-RV-102-04	5.54	2.79	2016-02-12	4.73	4.59	0.042	0.038	42.9	2022-10-18	
2-RV-102-05	5.54	2.79	2016-02-12	4.97	4.78	0.029	0.052	38.5	2022-10-18	
2-RV-102-06	5.54	2.79	2016-02-12	4.27	4.27	0.060	0.000	24.5	2022-10-18	

Line 3-RF-101 | Reflux | A312 TP316 | t_nom 5.49 mm | t-min 2.41 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RF-101-01	5.49	2.41	2016-02-12	5.31	5.40	0.003	-0.024	>50	2025-10-18	
3-RF-101-02	5.49	2.41	2016-02-12	5.34	5.38	0.003	-0.011	>50	2025-10-18	
3-RF-101-03	5.49	2.41	2016-02-12	5.33	5.35	0.004	-0.005	>50	2025-10-18	
3-RF-101-04	5.49	2.41	2016-02-12	5.35	5.29	0.007	0.016	>50	2025-10-18	
3-RF-101-05	5.49	2.41	2016-02-12	5.33	5.17	0.012	0.043	>50	2025-10-18	
3-RF-101-06	5.49	2.41	2016-02-12	5.23	5.12	0.017	0.030	>50	2025-10-18	
3-RF-101-07	5.49	2.41	2016-02-12	5.37	5.38	0.002	-0.003	>50	2025-10-18	
3-RF-101-08	5.49	2.41	—	—	5.23	—	—	—	2025-10-18	
3-RF-101-09	5.49	2.41	2016-02-12	5.19	5.13	-0.002	0.016	>50	2025-10-18	

Line 2-RF-102 | Reflux | A312 TP316 | t_nom 3.91 mm | t-min 2.16 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RF-102-01	3.91	2.16	2016-02-12	3.76	3.66	0.011	0.027	>50	2025-10-18	
2-RF-102-02	3.91	2.16	2016-02-12	3.64	3.65	0.010	-0.003	>50	2025-10-18	
2-RF-102-03	3.91	2.16	2016-02-12	3.75	3.68	0.011	0.019	>50	2025-10-18	
2-RF-102-04	3.91	2.16	2016-02-12	3.38	2.83	0.044	0.149	4.5	2022-01-14	RL/2 < interval
2-RF-102-05	3.91	2.16	2016-02-12	3.49	2.88	0.133	0.166	4.3	2021-12-19	RL/2 < interval
2-RF-102-06	3.91	2.16	2016-02-12	3.47	2.74	0.159	0.198	2.9	2021-04-04	RL/2 < interval
2-RF-102-07	3.91	2.16	2016-02-12	3.57	2.82	0.048	0.204	3.2	2021-06-01	RL/2 < interval

Line 4-LA-104 | L/R exchanger, lean side | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-104-01	6.02	3.18	2016-02-12	5.31	5.16	0.039	0.041	48.6	2025-10-18	
4-LA-104-02	6.02	3.18	2016-02-12	5.00	4.79	0.053	0.057	28.2	2025-10-18	
4-LA-104-03	6.02	3.18	2016-02-12	5.42	5.23	0.031	0.052	39.7	2025-10-18	
4-LA-104-04	6.02	3.18	2016-02-12	5.63	5.49	0.024	0.038	>50	2025-10-18	
4-LA-104-05	6.02	3.18	2016-02-12	4.88	4.71	0.056	0.046	27.2	2025-10-18	
4-LA-104-06	6.02	3.18	2016-02-12	5.55	5.38	0.030	0.046	47.6	2025-10-18	
4-LA-104-07	6.02	3.18	2016-02-12	5.28	5.15	0.036	0.035	>50	2025-10-18	
4-LA-104-08	6.02	3.18	2016-02-12	5.11	4.88	0.045	0.063	27.2	2025-10-18	
4-LA-104-09	6.02	3.18	—	—	4.61	—	—	—	2025-10-18	
4-LA-104-10	6.02	3.18	2016-02-12	4.84	4.63	0.057	0.057	25.4	2025-10-18	

Line 4-RA-104 | L/R exchanger, rich side | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-104-01	8.56	3.81	2016-02-12	7.71	7.15	0.058	0.152	21.9	2022-10-18	
4-RA-104-02	8.56	3.81	2016-02-12	7.85	7.31	0.053	0.147	23.8	2022-10-18	
4-RA-104-03	8.56	3.81	2016-02-12	8.28	7.94	0.071	0.092	44.7	2022-10-18	
4-RA-104-04	8.56	3.81	2016-02-12	8.18	7.64	0.097	0.147	26.1	2022-10-18	
4-RA-104-05	8.56	3.81	2016-02-12	8.16	7.57	0.095	0.160	23.5	2022-10-18	
4-RA-104-06	8.56	3.81	2016-02-12	7.61	7.13	0.062	0.130	25.5	2022-10-18	
4-RA-104-07	8.56	3.81	2016-02-12	7.32	6.67	0.081	0.177	16.2	2022-10-18	
4-RA-104-08	8.56	3.81	2016-02-12	7.81	7.46	0.083	0.095	38.4	2022-10-18	

Line 2-ST-101 | Reboiler steam | A106 Gr.B | t_nom 3.91 mm | t-min 2.29 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-ST-101-01	3.91	2.29	2016-02-12	3.61	3.59	0.014	0.005	>50	2025-10-18	
2-ST-101-02	3.91	2.29	2016-02-12	3.55	3.34	0.024	0.057	18.4	2025-10-18	
2-ST-101-03	3.91	2.29	2016-02-12	3.55	3.68	0.011	-0.035	>50	2025-10-18	
2-ST-101-04	3.91	2.29	2016-02-12	3.68	3.52	0.013	0.043	28.3	2025-10-18	
2-ST-101-05	3.91	2.29	2016-02-12	3.50	3.27	0.027	0.063	15.7	2025-10-18	
2-ST-101-06	3.91	2.29	2016-02-12	3.63	3.66	-0.008	-0.008	>50	2025-10-18	

Line 3-CW-101 | Amine cooler CW | A106 Gr.B | t_nom 5.49 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-CW-101-01	5.49	2.79	2016-02-12	5.03	4.91	0.022	0.033	>50	2025-10-18	
3-CW-101-02	5.49	2.79	2016-02-12	5.01	4.98	0.018	0.008	>50	2025-10-18	
3-CW-101-03	5.49	2.79	2016-02-12	4.93	4.87	0.028	0.016	>50	2025-10-18	
3-CW-101-04	5.49	2.79	2016-02-12	5.26	5.20	0.011	0.016	>50	2025-10-18	
3-CW-101-05	5.49	2.79	2016-02-12	5.35	5.14	0.014	0.057	41.2	2025-10-18	
3-CW-101-06	5.49	2.79	2016-02-12	5.08	5.01	0.019	0.019	>50	2025-10-18	

Line 3-CW-102 | Amine cooler CW | A106 Gr.B | t_nom 5.49 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-CW-102-01	5.49	2.79	2016-02-12	5.06	4.85	0.028	0.057	36.1	2025-10-18	
3-CW-102-02	5.49	2.79	2016-02-12	4.37	4.17	0.057	0.054	24.3	2025-10-18	
3-CW-102-03	5.49	2.79	2016-02-12	4.54	4.27	0.053	0.073	20.2	2025-10-18	
3-CW-102-04	5.49	2.79	2016-02-12	4.69	4.50	0.043	0.052	33.1	2025-10-18	
3-CW-102-05	5.49	2.79	2016-02-12	4.66	4.45	0.057	0.057	29.1	2025-10-18	

Sign-off	Name	Function	Date	Signature
Prepared	Trevor Nolan	UT technician, SNT-TC-1A UT Level II	2019-10-18	_____